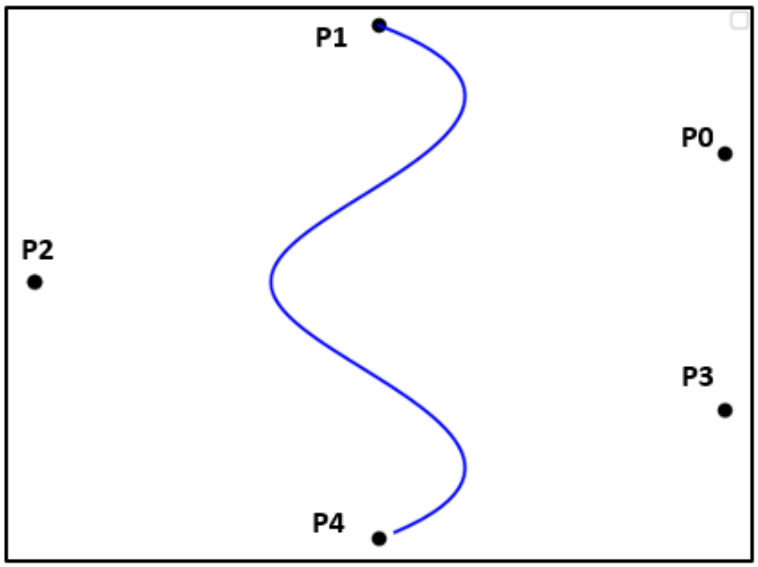
CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 1 | Marks 20 | **SET A**

Date: 27/7/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. **[6 marks]** Explain how Level of Detail (LoD) improves rendering performance in modern computer graphics. Mention one possible disadvantage.

2. **[14 marks]** A 2D Bezier curve Q is defined by the control points P0, P1, P2, P3, P4 (see the following figure), where the curve is symmetric about a line y = 3 and P2 is used twice. Compute the value of $Q(\frac{1}{2})$ if the vertices P0, P1 and P2 are (4, 7), (5, 5) and (3, 3).



CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 1 | Marks 20 | **SET B**

Date: 27/7/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. **[6 marks]** Describe the stages of the Graphics Pipeline. Explain why rasterization is placed before fragment shading.

2. **[14 marks]** Consider a 4x4 grayscale image IMG in the figure below. Apply the Haar Discrete Wavelet Transform to the image by performing a row-wise followed by column-wise transformation with a threshold value of 5.
   a. Find the resulting compressed DWT coefficient matrix and compression ratio if each pixel is stored using 8 bits.
   b. Reconstruct the image using inverse DWT and calculate the Mean Absolute Error (MAE) between the reconstructed image and the original.

| | | | |
|---|---|---|---|
| 12 | 16 | 24 | 28 |
| 14 | 18 | 26 | 30 |
| 30 | 26 | 18 | 14 |
| 32 | 28 | 20 | 16 |

CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 1 | Marks 20 | **SET C**

Date: 27/7/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. **[6 marks]** Explain why GPUs use different memory types. Compare their speed, accessibility and applications.

2. **[14 marks]** Consider a 3D Bezier curve Q, defined by control points (– 4, 4, 1), (– 2, 1, 3), (0, 1, 2), (2, 4, 5), (4, 5, 6) and (1, 3, 7). Compute the value of $Q(\frac{3}{5})$, if the curve has the same first point as the last point.

CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 1 | Marks 20 | **SET D**

Date: 27/7/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. **[6 marks]** Explain why B-Spline curves exhibit local control whereas Bézier curves exhibit global control.

2. **[14 marks]** Consider a 4x4 grayscale image IMG in the figure below where each pixel can hold up to 8 bits of data. Apply two compression methods on the image: Run-Length Encoding (RLE) and Haar Discrete Wavelet Transform (DWT) using a threshold value of 4. Determine which method gives better compression for this image. (Assume 1 byte is allocated to store the frequency count for each run.)

| | | | |
|---|---|---|---|
| 15 | 15 | 15 | 16 |
| 15 | 18 | 18 | 18 |
| 80 | 80 | 82 | 82 |
| 80 | 82 | 82 | 85 |

CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 1 | Marks 20 | **SET E**

Date: 27/7/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. **[6 marks]** Explain the working mechanism of OLED displays, including how the sub-pixels are oriented in it.
2. **[14 marks]** Consider a 3D Bezier curve Q, defined by control points (− 4, 4, 1), (− 2, 2, 3), (0, 3, 2), (2, 4, 5), (4, 5, 6) and (1, 3, 7). Compute the value of $Q(\frac{3}{5})$, if the curve has the same first point as the last point.

CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 1 | Marks 20 | **SET F**

Date: 27/7/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. **[6 marks]** Compare Uniform B-Spline and Open Uniform B-Spline curves. Explain why Open Uniform B-Spline curves are more widely used in CAD systems.

2. **[14 marks]** Consider a 4x4 grayscale image IMG in the figure below where each pixel can hold up to 8 bits of data. Apply two compression methods on the image: Run-Length Encoding (RLE) and Haar Discrete Wavelet Transform (DWT) using a threshold value of 4. Determine which method gives better compression for this image. (Assume 1 byte is allocated to store the frequency count for each run.)

| | | | |
|---|---|---|---|
| 20 | 20 | 22 | 24 |
| 20 | 22 | 24 | 26 |
| 60 | 60 | 62 | 64 |
| 60 | 62 | 64 | 66 |

CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 2 | Marks 20 | **SET A**

Date: 3/8/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. **[8 marks]** Two lines pass through the origin, one along the x-axis and the other making an angle φ with it, as shown in the figure below. Show that reflecting a point first about the x-axis and then about the second line is equivalent to a single rotation about the origin by angle 2φ.

2. **[12 marks]** Consider a rectangle OACB with vertices O(4, 4), A(4, 10), C(12, 10) and B(12, 4). Scale the rectangle by 150% along the x-direction and 200% along the y-direction, keeping point A fixed, as illustrated below. You must -

    a. Mention the steps.
    b. Determine the composite transformation matrix.
    c. Calculate and plot the final vertices.

CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 2 | Marks 20 | **SET B**

Date: 3/8/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. **[8 marks]** Show that the determinant of the composite transformation formed by applying a horizontal shear followed by a vertical shear is same as the product of the determinants of the individual horizontal and vertical shearing matrices.

2. **[12 marks]** Consider a square OACB with vertices O(2, 5), A(6, 5), C(6, 9) and B(2, 9). Reflect the square along the line y = -2x + 10 using a 2D transformation and find the final vertices.

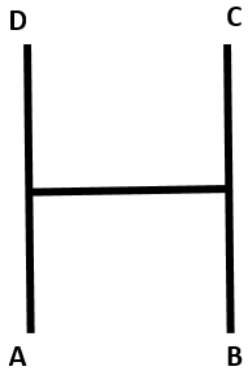
CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 2 | Marks 20 | **SET C**

Date: 3/8/2026 | Time: 30 minutes

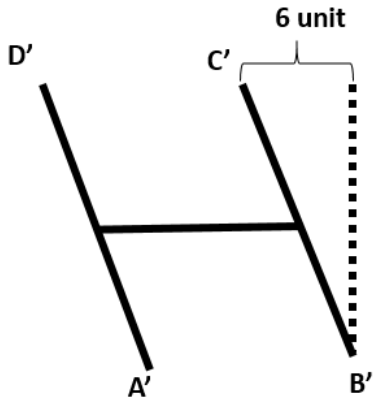
ID:..... Name:..... Section:.....

1. **[8 marks]** Prove that reflecting a point (x, 0, z) about the xy-plane is the same as a 180-degree rotation about the x-axis.

2. **[12 marks]** Consider a H-shaped object with vertices A(2, 4), B(6, 4), C(6, 12) and D(2, 12). Apply appropriate transformations on the object so that the vertices move 6 units to the left while keeping point A and B fixed.




(a) Initial State




(b) Transformed State

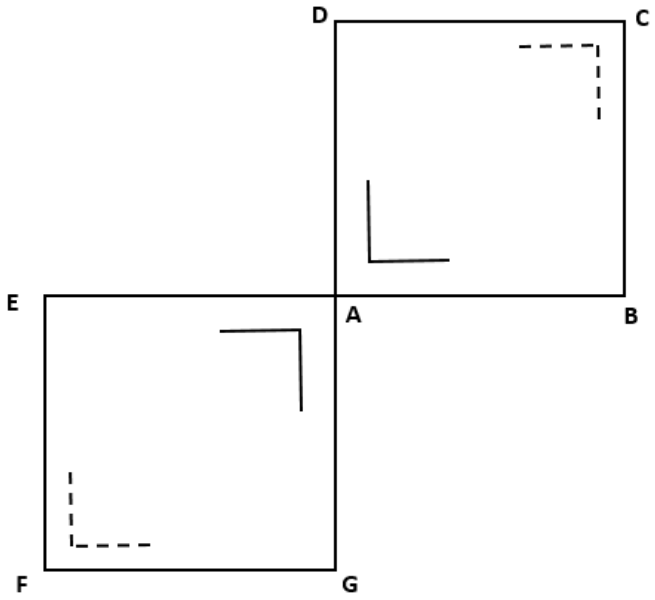
CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 2 | Marks 20 | **SET D**

Date: 3/8/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. **[8 marks]** Determine the composite transformation matrix required to map a 3D cube of size 2x2x2 and centered at (1,1,1) to a volume of size nx × ny x nz.

2. **[12 marks]** Consider a quadrilateral ABCD with vertices A(8, 4), B(16, 4), C(16, 10) and D(8, 10). Apply appropriate transformations to it to produce the quadrilateral AEFG with a shared vertex A. You must -
   a. Mention the steps.
   b. Determine the composite transformation matrix.
   c. Calculate the final vertices



CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 2 | Marks 20 | **SET E**

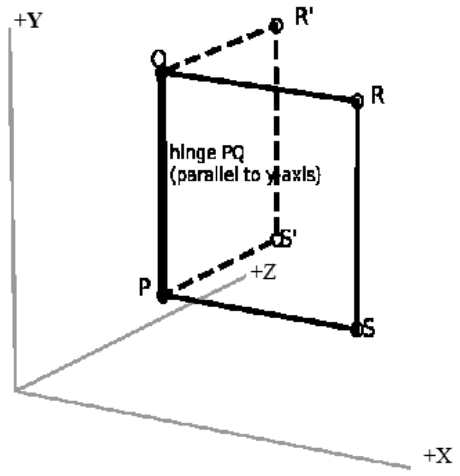
Date: 3/8/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. **[8 marks]** Prove that reflection about the line y = x is equivalent to interchanging the x- and y-coordinates of every point.

2. **[12 marks]** A door PQRS is hinged along its vertical edge PQ, which is parallel to the y-axis, with P(2, 2, 3), Q(2, 8, 3), R(7, 8, 3) and S(7, 2, 3), as shown in the figure below. Rotate the door by 60° about the hinge PQ, keeping P and Q fixed. You must —

   a. Mention the steps.
   b. Determine the composite transformation matrix.
   c. Calculate the final vertices R′ and S'.



CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 2 | Marks 20 | **SET F**

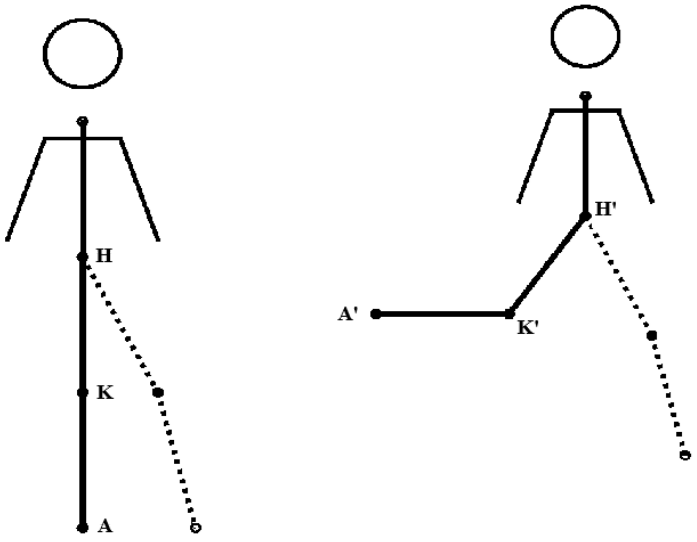
Date: 3/8/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. **[8 marks]** Prove that reflecting a point (x,y,0) about the xz-plane is equivalent to a 180-degree rotation about the z-axis.

2. **[12 marks]** A simplified skeleton character performs a running jump, as shown in the figure below. The character's hip is at H(5, 10), knee at K(5, 6), and ankle at A(5, 2), with the torso, arms, and head rigidly attached above the hip. During the jump: The entire character is translated 3 units upward and 2 units right (the jump itself), the thigh HK is then rotated 35° clockwise about the hip H and the shin KA is additionally rotated 55° clockwise about the knee. You must —

   a. Mention the steps to do the jump.
   b. Determine the composite transformation matrix for the thigh and for the shin.
   c. Calculate the final positions of H', K' and A'.




(a) Initial Pose (b) Running Pose

CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 3 | Marks 20 | **SET A**

Date: 31/8/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. Consider a viewport of size 720 × 480 in which the origin is placed at the top-left corner and the y-axis increases downward (instead of the standard bottom-left, y-axis-upward convention).

   **a. [6 Marks]** Identify and explain the rendering technique described by the given scenario.

   **b. [7 Marks]** Construct the modified viewport transformation matrix.

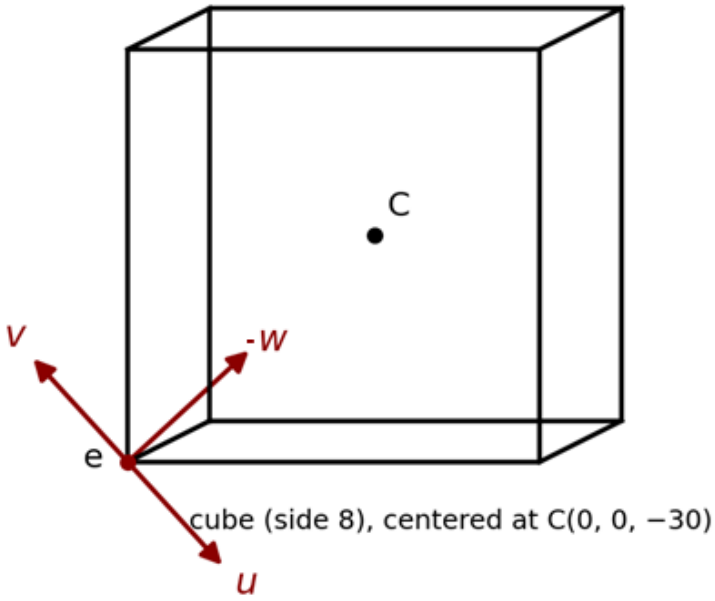
   **c. [7 Marks]** Transform point P(4, −2, −7) to this viewport. P lies within an orthographic view volume defined by l = −8, r = 8, b = −6, t = 6, n = −3, f = −12.

CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 3 | Marks 20 | **SET B**

Date: 31/8/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. A cube of side 8 units is centered at C(0, 0, −30) in the 3D canonical coordinate system (see figure below). A camera with origin e and basis {u, v, w} has its eye placed at the front-bottom-left corner of the cube, with the viewing direction pointing toward the cube's center C.
   a. **[6 marks]** Derive the perspective projection matrix to apply to a 1D model.
   b. **[7 marks]** Determine the basis vector of the camera for the given scenario.
   c. **[7 marks]** Find the position of point C with respect to the camera coordinate system.



CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 3 | Marks 20 | **SET C**

Date: 31/8/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. Consider a 3D scene where a line segment AB is defined in an orthographic view volume with l = −5, r = 5, b = −3, t = 9, n = −3, f = −9. Instead of mapping directly to the standard canonical view volume, the system first maps the coordinates to a normalized space defined by (x, y) ∈ [0, 4]. This normalized space is then mapped to a viewport of size 800 × 600, where the origin is at the bottom-left corner, with x increasing to the right and y increasing upward.

   **a. [6 Marks]** State three differences between image-order and object-order rendering.

   **b. [7 Marks]** Construct the matrix that maps the given orthographic view volume to the normalized [0, 4] space.

   **c. [7 Marks**] Apply the full transformation to determine the final screen coordinates of points A(−2, 5, −4) and B(3, −1, −8).

CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 3 | Marks 20 | **SET D**

Date: 31/8/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. Consider a line segment CD with vertices C(2, 1, −3) and D(9, −2, −8). The line undergoes a perspective projection with the following settings: Near plane = 2, Far plane = 9, Field-of-View (FoV) = 90°, Aspect ratio = 1:1. It is displayed on a viewport of size 500 × 500.
   a. **[6 Marks]** Illustrate the differences between perspective and orthographic projection.
   b. **[7 Marks]** Construct the perspective matrix P and the orthographic matrix $M_{orth}$.
   c. **[7 Marks]** Determine whether the line segment CD is fully visible inside the viewing frustum. Justify your answer.

CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 3 | Marks 20 | **SET E**

Date: 31/8/2026 | Time: 30 minutes

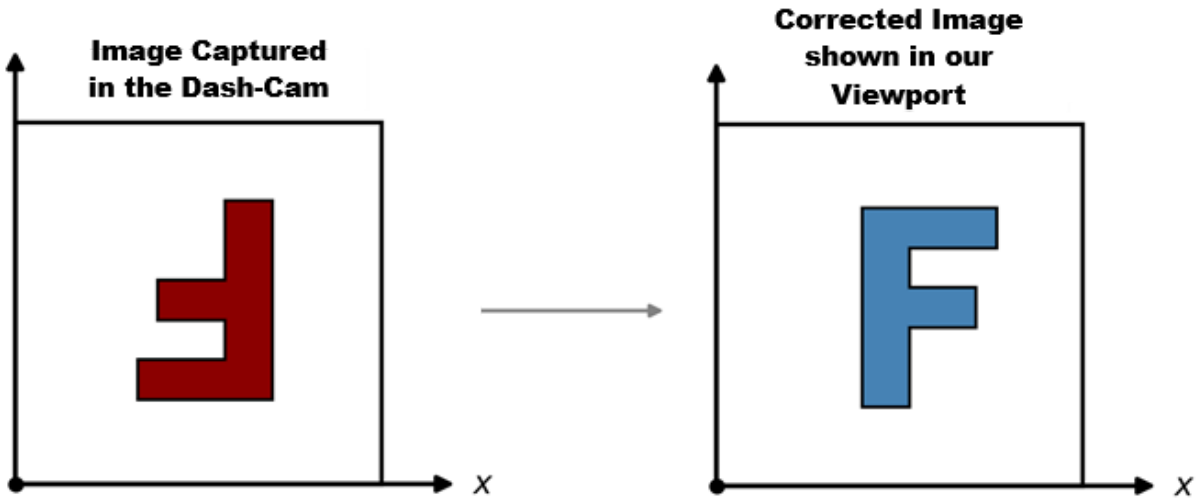
ID:..... Name:..... Section:.....

1. A rear-facing dash-cam is mounted so that its image comes out fully inverted. As shown in the figure below, the captured image from the dash-cam points to the opposite direction from the standard viewport. We need to build a modified viewport transformation matrix to show the inverted image to it's original form in a viewport of size 1024 × 780.

   **a. [6 Marks]** Describe different sequences of viewing transformation.

   **b. [7 Marks]** Construct the modified viewport transformation matrix for the given scenario.

   **c. [7 marks]** Transform the 3D line segment MN, with vertices M(1, 5, −3) and N(−4, −2, −7), from an orthographic view volume with l = −5, r = 5, b = −5, t = 5, n = −1, f = −8, to this viewport.



CSE 4203 | Computer Graphics | Fall 2025 | Quiz - 3 | Marks 20 | **SET F**

Date: 31/8/2026 | Time: 30 minutes

ID:..... Name:..... Section:.....

1. Here (in the figure below), origin O and basis vectors {z, y} construct a 2D canonical coordinate system, where −z is the viewing direction and y is the up direction. A circle of radius 5 is centered at point a. We want to view this scene from a new 2D camera with eye e and basis {w, v}, where e lies on the circle (as shown). The basis are obtained by rotating the canonical basis {z, y} by θ degrees, and the eye position is given by e = a + 5(cos θ, sin θ).

   **a. [6 Marks]** Show that the frame-to-canonical transformation is the inverse transformation of the canonical-to-frame transformation.

   **b. [7 Marks]** Determine the basis vectors.

   **c. [7 Marks]** Determine the position of point b with respect to the camera coordinate system.

